

Task 14: Linux Server Hardening & Secure Configuration

Tools Used :

- Ubuntu Linux
- SSH
- UFW Firewall
- Systemctl
- APT Package Manager
- Linux File Permission Utilities (chmod, ls)
- System Log Utilities (auth.log, journalctl)
- Lynis (Security Auditing Tool)

Introduction

In this task, I performed Linux server hardening using the Ubuntu operating system. The objective of this task was to secure the Linux system by identifying security risks, reducing system vulnerabilities, and applying secure configuration settings. Linux hardening helps in protecting systems from cyber attacks such as unauthorized access, brute force attacks, malware infections, and privilege misuse.

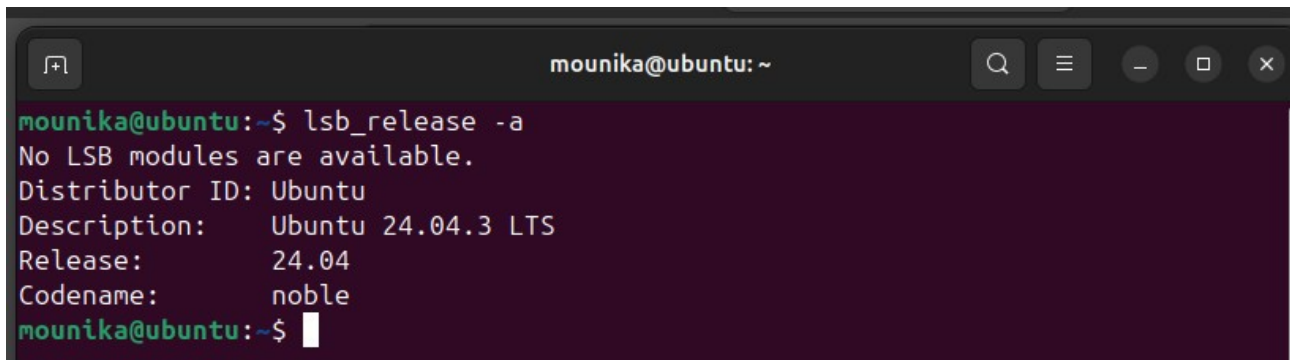
1. Review Default Linux System Settings

Initially, I reviewed the default system configuration to understand existing users, running services, and open network ports. I used system commands to check system exposure and active components.

I checked system users using the `/etc/passwd` file, which shows all user accounts available in the system. I also checked currently logged-in users and verified active system services. Additionally, I checked open ports to identify which network ports were accessible.

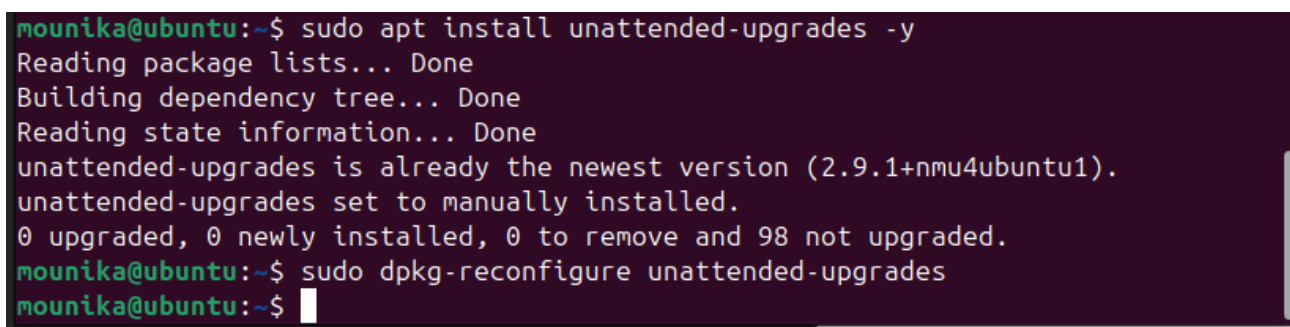
This step helped me understand the attack surface of the system.

Screenshot: Ubuntu version

A terminal window titled 'mounika@ubuntu: ~' with standard Ubuntu window controls. The terminal shows the command 'lsb_release -a' and its output: 'No LSB modules are available.', 'Distributor ID: Ubuntu', 'Description: Ubuntu 24.04.3 LTS', 'Release: 24.04', and 'Codename: noble'.

```
mounika@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 24.04.3 LTS
Release:      24.04
Codename:     noble
mounika@ubuntu:~$
```

Screenshot: Auto updates enabled

A terminal window titled 'mounika@ubuntu: ~' showing the installation of 'unattended-upgrades'. The output indicates the package is already the newest version and is set to manually installed. It also shows the command 'sudo dpkg-reconfigure unattended-upgrades' being entered.

```
mounika@ubuntu:~$ sudo apt install unattended-upgrades -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
unattended-upgrades set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 98 not upgraded.
mounika@ubuntu:~$ sudo dpkg-reconfigure unattended-upgrades
mounika@ubuntu:~$
```

Screenshot: List of system users

```

mounika@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987>User for polkitd:/:usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin

```

Screenshot: active login session

```

mounika@ubuntu:~$ who
mounika  seat0          2026-02-10 14:48 (login screen)
mounika  tty2           2026-02-10 14:48 (tty2)
mounika@ubuntu:~$

```

Screenshot: List of active services

```
mounika@ubuntu:~$ systemctl list-units --type=service
```

UNIT

```
accounts-daemon.service
alsa-restore.service
apache2.service
apparmor.service
apport.service
avahi-daemon.service
blk-availability.service
bluetooth.service
colord.service
console-setup.service
cron.service
cups-browsed.service
cups.service
dbus.service
fwupd.service
gdm.service
gnome-remote-desktop.service
kerneloops.service
keyboard-setup.service
kmod-static-nodes.service
lvm2-monitor.service
ModemManager.service
```

```
lines 1-23...skipping...
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
alsa-restore.service	loaded	active	exited	Save/Restore Sound Card State
apache2.service	loaded	active	running	The Apache HTTP Server
apparmor.service	loaded	active	exited	Load AppArmor profiles
apport.service	loaded	active	exited	automatic crash report generation
avahi-daemon.service	loaded	active	running	Avahi mDNS/DNS-SD Stack
blk-availability.service	loaded	active	exited	Availability of block devices
bluetooth.service	loaded	active	running	Bluetooth service
colord.service	loaded	active	running	Manage, Install and Generate Color Profiles
console-setup.service	loaded	active	exited	Set console font and keymap
cron.service	loaded	active	running	Regular background program processing daemon
cups-browsed.service	loaded	active	running	Make remote CUPS printers available locally
cups.service	loaded	active	running	CUPS Scheduler
dbus.service	loaded	active	running	D-Bus System Message Bus
fwupd.service	loaded	active	running	Firmware update daemon
gdm.service	loaded	active	running	GNOME Display Manager
gnome-remote-desktop.service	loaded	active	running	GNOME Remote Desktop
kerneloops.service	loaded	active	running	Tool to automatically collect and submit kernel crash signatures
keyboard-setup.service	loaded	active	exited	Set the console keyboard layout
kmod-static-nodes.service	loaded	active	exited	Create List of Static Device Nodes
lvm2-monitor.service	loaded	active	exited	Monitoring of LVM2 mirrors, snapshots etc. using dmeventd or progress
ModemManager.service	loaded	active	running	Modem Manager
mysql.service	loaded	active	running	MySQL Community Server
NetworkManager-wait-online.service	loaded	active	exited	Network Manager Wait Online
NetworkManager.service	loaded	active	running	Network Manager
openvpn.service	loaded	active	exited	OpenVPN service
packagekit.service	loaded	active	running	PackageKit Daemon
plymouth-quit-wait.service	loaded	active	exited	Hold until boot process finishes up
plymouth-read-write.service	loaded	active	exited	Tell Plymouth To Write Out Runtime Data
plymouth-start.service	loaded	active	exited	Show Plymouth Boot Screen
polkit.service	loaded	active	running	Authorization Manager
power-profiles-daemon.service	loaded	active	running	Power Profiles daemon
rsyslog.service	loaded	active	running	System Logging Service
rtkit-daemon.service	loaded	active	running	RealtimeKit Scheduling Policy Service
setvtrgb.service	loaded	active	exited	Set console scheme
snapp-apparmor.service	loaded	active	exited	Load AppArmor profiles managed internally by snapp
snapp-seeded.service	loaded	active	exited	Wait until snapp is fully seeded
snapp.service	loaded	active	running	Snapp Daemon
ssh.service	loaded	active	running	OpenBSD Secure Shell server
switcheroo-control.service	loaded	active	running	Switcheroo Control Proxy service
sysstat.service	loaded	active	exited	Resets System Activity Logs
systemd-backlight@backlight:intel_backlight.service	loaded	active	exited	Load/Save Screen Backlight Brightness of backlight:intel_backlight

Screenshot: Ports + LISTEN status

```
mounika@ubuntu:~$ sudo ss -tuln
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
udp    UNCONN 0        0        0.0.0.0:55863       0.0.0.0:*
udp    UNCONN 0        0        127.0.0.54:53       0.0.0.0:*
udp    UNCONN 0        0        127.0.0.53%lo:53    0.0.0.0:*
udp    UNCONN 0        0        0.0.0.0:5353        0.0.0.0:*
udp    UNCONN 0        0        0.0.0.0:54523       0.0.0.0:*
udp    UNCONN 0        0        [::]:48062          [::]:*
udp    UNCONN 0        0        [::]:5353           [::]:*
tcp    LISTEN 0        151      127.0.0.1:3306       0.0.0.0:*
tcp    LISTEN 0        4096     127.0.0.53%lo:53    0.0.0.0:*
tcp    LISTEN 0        4096     127.0.0.54:53       0.0.0.0:*
tcp    LISTEN 0        4096     0.0.0.0:22          0.0.0.0:*
tcp    LISTEN 0        4096     127.0.0.1:631       0.0.0.0:*
tcp    LISTEN 0        70       127.0.0.1:33060     0.0.0.0:*
tcp    LISTEN 0        4096     [::1]:631           [::]:*
tcp    LISTEN 0        4096     [::]:22             [::]:*
tcp    LISTEN 0        511      *:80                *:*
```

2. User Account and Sudo Access Management

In this step, I reviewed system users and ensured only required users had administrative access. I verified sudo privileges using group and sudo configuration checks. This step follows the principle of least privilege, which means users should have only the access required for their role.

Restricting unnecessary sudo access helps prevent privilege escalation and insider threats.

Screenshot: User in sudo group OR sudo config

```
mounika@ubuntu:~$ groups $USER
mounika : mounika adm cdrom sudo dip plugdev users lpadmin wireshark vboxusers
mounika@ubuntu:~$ sudo visudo
visudo: /etc/sudoers.tmp unchanged
mounika@ubuntu:~$
```

3. SSH Security Hardening

I secured remote login by modifying the SSH configuration file. I disabled root login and password authentication and ensured key-based authentication is enabled.

Changes applied:

- Disabled root login

- Disabled password-based login
- Enabled public key authentication

This helps prevent brute force attacks and unauthorized root access attempts.

4. System Update and Patch Management

I updated all system packages using the package manager and enabled automatic security updates. Regular system updates help fix known vulnerabilities and keep the system protected against newly discovered threats.

Automatic updates ensure the system remains secure without requiring manual intervention.

Screenshot : Change / Confirm These Values

```
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem        sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#      X11Forwarding no
#      AllowTcpForwarding no
#      PermitTTY no
#      ForceCommand cvs server

PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
```

```
Cannot parse dst/src address.
mounika@ubuntu:~$ sudo systemctl restart ssh
mounika@ubuntu:~$ sudo sshd -T | grep -E "permitrootlogin|passwordauthentication
|pubkeyauthentication"
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
mounika@ubuntu:~$
```


5. Firewall Configuration

I configured the Uncomplicated Firewall (UFW) and allowed only required network ports. Firewall acts as a protective barrier that blocks unauthorized incoming traffic while allowing legitimate traffic.

I enabled firewall and allowed SSH port to maintain remote connectivity.

Enable Firewall

```
mounika@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
```



```
mounika@ubuntu:~$ sudo ufw allow 22
Skipping adding existing rule
Skipping adding existing rule (v6)
mounika@ubuntu:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
22	ALLOW	Anywhere
80	ALLOW	Anywhere
443	ALLOW	Anywhere
23	DENY	Anywhere
Anywhere	DENY	192.168.1.100
22 (v6)	ALLOW	Anywhere (v6)
80 (v6)	ALLOW	Anywhere (v6)
443 (v6)	ALLOW	Anywhere (v6)
23 (v6)	DENY	Anywhere (v6)

```
mounika@ubuntu:~$
```

6. Service Hardening

I reviewed running services and disabled unnecessary services. Each running service increases attack surface, so disabling unused services improves system security.

Screenshots : service list

```
mounika@ubuntu:~$ systemctl list-unit-files --type=service
```

UNIT FILE	STATE	PRESET
accounts-daemon.service	enabled	enabled
alsa-restore.service	static	-
alsa-state.service	static	-
alsa-utils.service	masked	enabled
anacron.service	enabled	enabled
apache-htcacheclean.service	disabled	enabled
apache-htcacheclean@.service	disabled	enabled
apache2.service	enabled	enabled
apache2@.service	disabled	enabled
apparmor.service	enabled	enabled
apport-autoreport.service	static	-
apport-coredump-hook@.service	static	-
apport-forward@.service	static	-
apport.service	enabled	enabled
apt-daily-upgrade.service	static	-
apt-daily.service	static	-
apt-news.service	static	-
autovt@.service	alias	-
avahi-daemon.service	enabled	enabled
blk-availability.service	enabled	enabled
bluetooth.service	enabled	enabled
bolt.service	static	-

```
lines 1-23...skipping...
```

Screenshot : Service disabled

```
mounika@ubuntu:~$ sudo systemctl stop bluetooth
mounika@ubuntu:~$ sudo systemctl disable bluetooth
Synchronizing state of bluetooth.service with SysV service script with /usr/lib/
systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable bluetooth
Removed "/etc/systemd/system/bluetooth.target.wants/bluetooth.service".
Removed "/etc/systemd/system/dbus-org.bluez.service".
mounika@ubuntu:~$
```

7. File Permission Security

I verified permissions of sensitive system files such as `/etc/shadow`, which stores password hashes. Ensuring correct file permissions prevents unauthorized access and protects user credentials.

Secure File Permissions

```
mounika@ubuntu:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1379 Jan 30 11:46 /etc/shadow
mounika@ubuntu:~$
```


8. Log Monitoring

I reviewed authentication logs to monitor login attempts and system activity. Log monitoring helps detect suspicious activities and possible intrusion attempts.

Check Authentication Logs

```
mounika@ubuntu:~$ sudo tail /var/log/auth.log
2026-02-10T15:23:37.876450+05:30 ubuntu dbus-daemon[1214]: [system] Rejected send message, 0 matched rules; type="error", sender=":1.73" (uid=1000 pid=2701 comm="/usr/bin/wireplumber" label="unconfined") interface="(unset)" member="(unset)" error name="org.bluez.Profile1.Error.NotImplemented" requested_reply="0" destination=":1.8" (uid=0 pid=1213 comm="/usr/libexec/bluetooth/bluetoothd" label="unconfined")
2026-02-10T15:23:37.876579+05:30 ubuntu dbus-daemon[1214]: [system] Rejected send message, 0 matched rules; type="error", sender=":1.73" (uid=1000 pid=2701 comm="/usr/bin/wireplumber" label="unconfined") interface="(unset)" member="(unset)" error name="org.bluez.Profile1.Error.NotImplemented" requested_reply="0" destination=":1.8" (uid=0 pid=1213 comm="/usr/libexec/bluetooth/bluetoothd" label="unconfined")
2026-02-10T15:23:37.889524+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-10T15:23:44.386077+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/systemctl disable bluetooth
2026-02-10T15:23:44.387108+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-10T15:23:45.524507+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-10T15:25:01.232887+05:30 ubuntu CRON[8529]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-02-10T15:25:01.237199+05:30 ubuntu CRON[8529]: pam_unix(cron:session): session closed for user root
2026-02-10T15:25:39.630350+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log
2026-02-10T15:25:39.631049+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
mounika@ubuntu:~$
```

9. Security Audit Using Lynis

I performed a system security audit using the Lynis security auditing tool. Lynis scans the system and provides security warnings, suggestions, and hardening recommendations.

From the scan results, I observed:

- System hardening status
- Security warnings (such as system reboot recommendation)
- Security improvement suggestions

This helped me understand additional security improvements needed for the system.

Hardening Section

```
--
2026-02-10 15:46:35 Hardening index : [67] [#####]
2026-02-10 15:46:35 Hardening strength: System has been hardened, but could use additional hardening
2026-02-10 15:46:35 ===
2026-02-10 15:46:37 Tool tips: enabled
2026-02-10 15:46:37 =====
2026-02-10 15:46:37 Tests performed:      288
2026-02-10 15:46:37 Total tests:         476
2026-02-10 15:46:37 Active plugins:       2
2026-02-10 15:46:37 Total plugins:       2
2026-02-10 15:46:37 =====
2026-02-10 15:46:37 Lynis 3.1.6
2026-02-10 15:46:37 2007-2025, CISOfy - https://cisofy.com/lynis/
mounika@ubuntu:~/lynis$
```

Warnings section

```
mounika@ubuntu:~/lynis$ sudo grep -i warning /var/log/lynis.log -A 5
2026-02-10 15:46:31 Skipped test CONT-8104 (Checking Docker info for any warnings)
2026-02-10 15:46:31 Reason to skip: Prerequisites not met (ie missing tool, other type of Linux distribution)
2026-02-10 15:46:31 ====
2026-02-10 15:46:31 Skipped test CONT-8106 (Gather basic stats from Docker)
2026-02-10 15:46:31 Reason to skip: Prerequisites not met (ie missing tool, other type of Linux distribution)
2026-02-10 15:46:31 ====
mounika@ubuntu:~/lynis$
```

Hardening Index

```
mounika@ubuntu:~/lynis$ sudo grep -i "hardening index" /var/log/lynis.log
2026-02-10 15:46:35 Hardening index : [67] [#####]
mounika@ubuntu:~/lynis$
```

Conclusion

Linux server hardening is an essential process to secure systems from common cyber threats and unauthorized access. By applying security configurations such as restricting user privileges, securing SSH access, enabling firewall protection, updating system packages, managing services, and securing file permissions, the overall attack surface of the system was significantly reduced. Log monitoring and security auditing further helped in identifying potential risks and ensuring the system follows secure configuration standards. Proper Linux hardening ensures system stability, data protection, and improved resistance against security attacks in real-world environments.